

FORWARD AND INVERSE KINEMATICS SOLUTION FOR 6-DOF MANIPULATORS WITH OFFSETS AND SPHERICAL WRISTS

ABSTRACT

This research presents a comprehensive, analytical solution for the forward and inverse kinematics of a 6-DOF industrial robot, specifically focusing on a common configuration that includes both joint offsets and a spherical wrist. The forward kinematics solution is established using the standard Denavit-Hartenberg (DH) convention. For the inverse kinematics, a geometric solution is employed, which is chosen for its intuitive and interpretable nature. This method uses kinematic decoupling, solving the problem by dividing the manipulator into two 3-DOF mechanisms: the first three joints for positioning the wrist center, and the final three spherical wrist joints for orientation.

The primary contribution of this work is the validation of the geometric decoupling method against the forward kinematic model, demonstrating that the analytical approach yields exact solutions without the iterative errors inherent in numerical solvers.

1. INTRODUCTION

Industrial robotic manipulators are a critical component in modern automation, with 6-DOF (Degrees of Freedom) serial-chain arms, such as those from KUKA, representing common and versatile morphology. The effective implementation of any control system, motion planner, or simulation for these robots is fundamentally dependent on a precise and computationally efficient kinematic model. Kinematics, the description of motion without consideration of the forces and torques that cause it, provides the essential mapping between the robot's configuration in joint space and the pose of its end-effector in the operational task-space.

This paper presents the complete forward and inverse kinematic analysis of a custom-designed 6-DOF manipulator. The objective is to derive a robust mathematical model that serves as the foundation for future control and trajectory planning tasks.

The analysis is performed in a sequential manner. First, the robot's structure is parameterized using the **Standard Denavit-Hartenberg (DH) convention**. This process involves the systematic assignment of coordinate frames to each of the robot's six joints to derive a concise set of parameters (a, i, θ, α) that define the manipulator's geometry.

Second, these DH parameters are used to solve the **Forward Kinematics (FK)** problem. By developing the homogeneous transformation matrices for each link, the composite transformation from the robot's base frame to the end-effector frame is formulated. This allows for the direct computation of the end-effector's position and orientation for any given set of joint angles.

Finally, the **Inverse Kinematics (IK)** problem is addressed. This involves determining the set of joint angles $(q_1 \dots q_2)$ required to position the end-effector at a specific, desired pose (position and orientation) in its workspace.

1. Algebraic Solution

Solves the non-linear equations from the forward kinematics transformation $T_n^0(q) = T_{desired}$.

- **Advantages:** Robust and efficient for real-time use when a closed-form solution exists; independent of the DH convention; effective for simple manipulators (≤ 3 DOF).
- **Limitations:** Poor scalability—complexity grows exponentially with DOF and link offsets, making it impractical for most 6-DOF manipulators.

2. Geometric Solution

Decomposes the manipulator's geometry into 2D trigonometric problems using the DH convention.

- **Advantages:** Fast and deterministic; widely used for 6-DOF manipulators (especially with spherical wrists); compatible with standard DH modeling.
- **Limitations:** Produces multiple valid solutions (e.g., elbow-up/down); must handle kinematic singularities where DOFs are lost.

3. Quaternion Solution

Uses 4D quaternions to represent and solve rotations.

- **Advantages:** Compact formulation; avoids gimbal lock; fewer equations.
- **Limitations:** Mathematically less intuitive; mainly suited for rotational systems; results less physically interpretable.

The resulting kinematic model provides the necessary foundation for all higher-level tasks, including singularity analysis, trajectory planning, and the future development of dynamic controllers for the manipulator.

Given that the kinematic structure under study involves rotational motion for each degree of freedom (DOF), employing a quaternion-based formulation would provide the most mathematically robust representation. However, for the purposes of clarity and intuitive understanding, this paper adopts a geometric solution approach. Although the robot joints exist in three-dimensional space, they are projected onto a single plane to simplify the analytical process. The spatial separations between consecutive joints are referred to as *offsets*. The exemplified manipulator features two such offsets: one located between the first and second joints, and another between the third and fourth joints.



Figure 1. An example robot CAD model

The fourth, fifth, and sixth joints collectively constitute a spherical wrist configuration. Accordingly, the six-degree-of-freedom (6-DOF) manipulator is decomposed into two subsystems: the first comprising joints one through three, and the second encompassing the spherical wrist assembly. It is important to note that the inverse kinematics (IK) of such systems inherently yield multiple possible joint configurations for a given end-effector pose. This phenomenon has been extensively addressed in prior research. In this study, only a single representative solution is presented—specifically, the one that provides the broadest and most continuous range of motion across the manipulator's workspace.

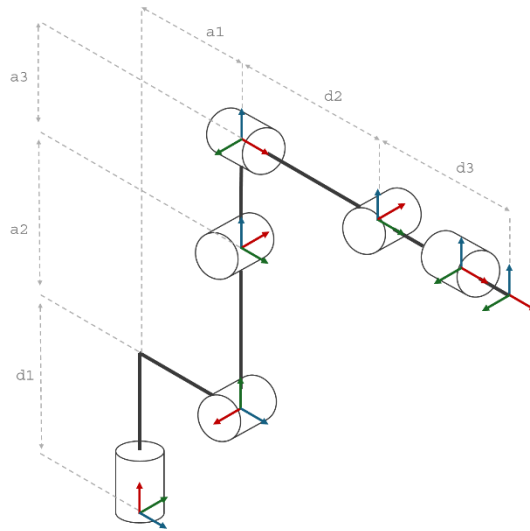


Figure 2. Schematic of robot shown at Figure 1

2. ROBOT VARIABLES, PARAMETERS & DENAVIT HARTENBERG TABLE

A very commonly used 6-DOF robot has the following variables, parameters, parameters and corresponding Denavit-Hartenberg (DH) table. In this section, these known and unknown will be exposed using the schematics in previous section.

2.1 Variables

The 6-DOF robot manipulator consists of six revolute joints, each providing one rotational degree of freedom. By adjusting the rotation of these joints, the robot can position its end-effector to perform the desired tasks. The primary kinematic variables of the system are the six revolute joint angles, denoted as θ . At the home position, all joint angles are defined as 0° , and during operation each angle must remain within the range of -180° to $+180^\circ$.

Through this paper and its supplementary materials, the following notation is used:

Subscript	Description
θ	Joint angle
$n = 1 \dots 6$	Indicates the joint number, counted from the base (joint 1) to the end-effector (joint 6)
v	Variable joint angle (active rotation from home position)
h	Home offset angle at the home position
f	Joint angle used in forward kinematics (includes both variable and home offset components)
i	Joint angle used in inverse kinematics (includes both variable and home offset components)

Table 1: Table of Subscripts

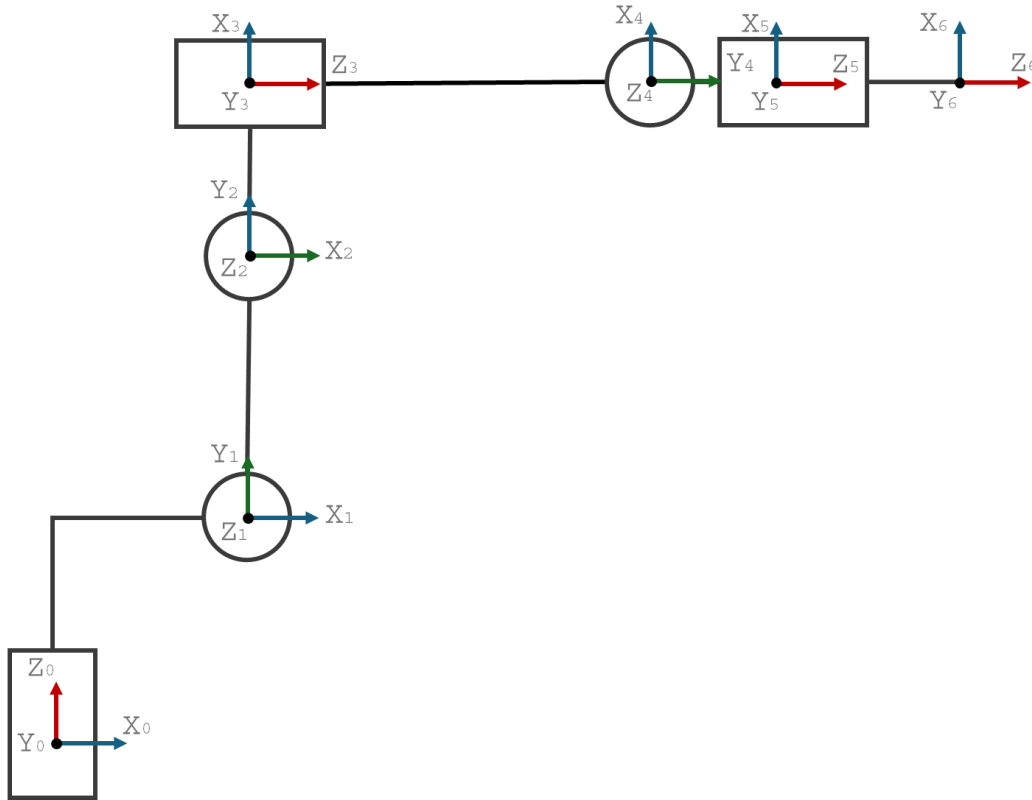


Figure 3: 2D Representation of schematics on Figure 2

2.2 Parameters & Denavit-Hartenberg Table

For the specified home position illustrated in **Figures 2 and 3**, the robot's geometry is defined using the Denavit–Hartenberg (DH) convention. In this representation, each link and joint of the 6-DOF manipulator is described by four parameters: **joint angle (θ)**, **link twist (α)**, **link offset (d)**, and **link length (a)**. Each row of the DH table corresponds to one joint, resulting in a 6×4 table that fully characterizes the manipulator's kinematic chain.

The definitions of the DH parameters are as follows:

- θ — Angle about the previous Z-axis, measured from the old X-axis to the new X-axis.
- α — Angle about the new X-axis, measured from the old Z-axis to the new Z-axis.
- d — Offset distance along the previous Z-axis, from the old joint to the new joint center.
- a — Length along the new X-axis, from the old joint to the new joint center.

When assigning local coordinate frames, the following conventions are applied:

- The **Z-axis** coincides with the joint axis — rotational for revolute joints or translational for prismatic joints.
- The **X-axis** is chosen to be perpendicular to both the previous and current Z-axes, intersecting them when possible.
- The **Y-axis** is determined according to the right-hand rule based on the orientations of the X and Z axes.

By following these conventions and using the geometric constraints illustrated in Figures 2 and 3, the complete DH parameter set of the robot manipulator (summarized in **Table 1**) can be derived. Proper placement and orientation of the local coordinate frames are critical steps in constructing the DH table, as they directly affect the simplicity and clarity of the kinematic model.

$link(i)$	θ_i	d_i	a_i	α_i
1	θ_1	$d1$	$a1$	90°
2	$\theta_2 + 90^\circ$	0	$a2$	-180°
3	θ_3	0	$a3$	-90°
4	θ_4	$d2$	0	90°
5	θ_5	0	0	-90°
6	θ_6	$d3$	0	0°

Table 2: D-H Parameters of robot manipulator that's schematics shown in Figure 2&3

In summary, the forward kinematics of the manipulator are determined by treating each joint's rotation angle as a variable. Since every joint is characterized by four Denavit–Hartenberg (DH) parameters, a 6-DOF robot has six joint variables and a total of $6 \times 4 = 24$ DH parameters that collectively define the end-effector's position in Cartesian space (x, y, z) and its three orientation angles.

For forward kinematics analysis, these six joint variables always remain the fundamental unknowns, regardless of whether the mechanism has 2, 3, or 10 degrees of freedom. Increasing the number of DOF has minimal impact on the forward kinematics process; however, the same cannot be said for inverse kinematics. As the number of joints and their configurations increase, the complexity of solving the inverse kinematics problem also grows, and certain analytical methods may no longer be applicable.

3. FORWARD KINEMATICS CALCULATION

In 1955, Denavit and Hartenberg introduced a matrix-based approach for systematically assigning coordinate frames to each link in a robotic manipulator. This method enables the precise description of both translational and rotational relationships between consecutive links. The kinematic model of the robot in this study is formulated according to the Denavit–Hartenberg (D-H) convention. The transformation between any two successive joints can be expressed by substituting the corresponding D-H parameters from the table into the transformation matrix, denoted as A_n , where:

$$A_{nf} = \begin{bmatrix} \cos \theta_{nf} & -\sin \theta_{nf} \cos \alpha_{nf} & \sin \theta_{nf} \sin \alpha_{nf} & a_{nf} \cos \theta_{nf} \\ \sin \theta_{nf} & \cos \theta_{nf} \cos \alpha_{nf} & \cos \theta_{nf} \sin \alpha_{nf} & a_{nf} \sin \theta_{nf} \\ 0 & \sin \alpha_{nf} & \cos \alpha_{nf} & d_{nf} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

When known joint parameters are put into their corresponding place in this matrix, then the following matrices are obtained for each joint starting from the base (1) to the tip (6);

$$A_{1f} = \begin{bmatrix} \cos \theta_{1f} & 0 & \sin \theta_{1f} & a_{1f} \cos \theta_{1f} \\ \sin \theta_{1f} & 0 & -\cos \theta_{1f} & a_{1f} \sin \theta_{1f} \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{2f} = \begin{bmatrix} \cos \theta_{2f} & \sin \theta_{2f} & 0 & a_{2f} \cos \theta_{2f} \\ \sin \theta_{2f} & \cos \theta_{2f} & 0 & a_{2f} \sin \theta_{2f} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{3f} = \begin{bmatrix} \cos \theta_{3f} & 0 & -\sin \theta_{3f} & a_{3f} \cos \theta_{3f} \\ \sin \theta_{3f} & 0 & \cos \theta_{3f} & a_{3f} \sin \theta_{3f} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{4f} = \begin{bmatrix} \cos \theta_{4f} & 0 & \sin \theta_{4f} & 0 \\ \sin \theta_{4f} & 0 & -\cos \theta_{4f} & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{5f} = \begin{bmatrix} \cos \theta_{5f} & 0 & -\sin \theta_{5f} & 0 \\ \sin \theta_{5f} & 0 & \cos \theta_{5f} & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{6f} = \begin{bmatrix} \cos \theta_{6f} & -\sin \theta_{6f} & 0 & 0 \\ \sin \theta_{6f} & \cos \theta_{6f} & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

For abbreviation and simplicity following notation substitutions will be used throughout this article.

$$C_n = \cos \theta_n$$

$$S_n = \sin \theta_n$$

$$C_{ab} = C_a C_b - S_a S_b$$

$$S_{ab} = C_a C_b + S_a S_b$$

The total transformation matrix from the robot base to the hand is as follows:

$$T_f = A_{1f}A_{2f}A_{3f}A_{4f}A_{5f}A_{6f} = \begin{bmatrix} r_{11f} & r_{12f} & r_{13f} & p_{xf} \\ r_{21f} & r_{22f} & r_{23f} & p_{yf} \\ r_{31f} & r_{32f} & r_{33f} & p_{zf} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

So:

- $r_{11f} = ((s_1s_4 + c_1c_4C_{2-3})c_5 + s_5s_{2-3}c_1)c_6 + (s_1c_4 - s_4c_1C_{2-3})s_6$
- $r_{12f} = -((s_1s_4 + c_1c_4C_{2-3})c_5 + s_5s_{2-3}c_1)s_6 + (s_1c_4 - s_4c_1C_{2-3})c_6$
- $r_{13f} = -(s_1s_4 + c_1c_4C_{2-3})s_5 + s_{2-3}c_1c_5$
- $r_{21f} = -((-s_1c_4C_{2-3} + s_4c_1)c_5 - s_1s_5s_{2-3})c_6 - (s_1s_4C_{2-3} + c_1c_4)s_6$
- $r_{22f} = ((-s_1c_4C_{2-3} + s_4c_1)c_5 - s_1s_5s_{2-3})s_6 - (s_1s_4C_{2-3} + c_1c_4)c_6$
- $r_{23f} = (-s_1c_4C_{2-3} + s_4c_1)s_5 + s_1s_{2-3}c_5$
- $r_{31f} = -(s_5C_{2-3} - s_{2-3}c_4c_5)c_6 - s_4s_6s_{2-3}$
- $r_{32f} = (s_5C_{2-3} - s_{2-3}c_4c_5)s_6 - s_4s_{2-3}c_6$
- $r_{33f} = -s_5s_{2-3}c_4 - c_5C_{2-3}$
- $p_{xf} = a_1c_1 + a_2c_1c_2 + a_3c_1C_{2-3} + d_2s_{2-3}c_1 - d_3s_1s_4s_5 - d_3s_5c_1c_4C_{2-3} + d_3s_{2-3}c_1c_5$
- $p_{yf} = a_1s_1 + a_2s_1c_2 + a_3s_1C_{2-3} + d_2s_1s_{2-3} - d_3s_1s_5c_4C_{2-3} + d_3s_1s_{2-3}c_5 + d_3s_4s_5c_1$
- $p_{zf} = a_2s_2 + a_3s_{2-3} + d_1 - d_2C_{2-3} - d_3s_5s_{2-3}c_4 - d_3c_5C_{2-3}$

The forward kinematics solution yields a 4 x 4 homogeneous transformation matrix, T_f . The upper-left 3 x 3 submatrix, T_{Rf} , is the rotation matrix

$$T_{Rf} = \begin{bmatrix} r_{11f} & r_{12f} & r_{13f} \\ r_{21f} & r_{22f} & r_{23f} \\ r_{31f} & r_{32f} & r_{33f} \end{bmatrix}$$

This is the Vectorial Representation (The Rotation Matrix) , This matrix describes the orientation of the end-effector frame relative to the base frame by explicitly defining the end-effector's basis vectors (n, o, a) in base coordinates:

- The $\vec{n}$ vector $[r_{11f}r_{12f}r_{13f}]^T$: This is the X-axis of your robot's hand (the "normal" direction), as seen from the base
- The $\vec{o}$ vector $[r_{21f}r_{22f}r_{23f}]^T$: This is the Y-axis of your robot's hand (the "orientation" or side-to-side direction), as seen from the base
- The $\vec{a}$ vector $[r_{31f}r_{32f}r_{33f}]^T$: This is the Z-axis of your robot's hand (the "approach" direction), as seen from the robot's base².

While mathematically complete, this is a non-minimal and non-intuitive way to specify a desired orientation. The nine constrained values do not correspond to simple, independent rotations, making it difficult for a user to define a target.

The unit vector shows the direction of the X, Y, Z-axis at the robot tip according to the base coordinate system. However, the expectation is to express all of these in angular form rather than vectorial. Below find how to express tip rotation in ZY'Z'' Euler and XY'Z'' Tait-Bryan angles. The matrices used to calculate these angles are available from.

ZY'Z'' Euler angles of the robot tip coordinate axis to the base coordinate axis (subscript e denotes Euler);

$$\begin{aligned} Z_{ef} &= \arctan2\left(\frac{a_{yf}}{a_{xf}}\right) \\ Y'_{ef} &= \arctan2\left(\frac{\sqrt{1 - a_{zf}^2}}{a_{zf}}\right) \\ Z''_{ef} &= \arctan2\left(\frac{a_{zf}}{-n_{zf}}\right) \end{aligned}$$

The translation matrix from the base coordinate system of the robot to the tip of the robot:

$$T_{Tf} = \begin{bmatrix} p_{xf} \\ p_{yf} \\ p_{zf} \end{bmatrix}$$

The transformation matrix that both includes translation and ZY'Z'' Euler angles can be represented as

$$T_{ef} = \begin{bmatrix} p_{xf} \\ p_{yf} \\ p_{zf} \\ Z_{ef} \\ Y'_{ef} \\ Z''_{ef} \end{bmatrix}$$

This concludes the forward kinematics calculation. In proceeding inverse kinematics calculations derived above T_{ef} matrix will be taken as input and θn joint rotations will be derived.

4. INVERSE KINEMATICS CALCULATIONS

In the previous section, the end-effector pose in Cartesian space was obtained by substituting specific joint angles into the forward kinematics model. Industrial robots, however, operate in the opposite direction. A robot controller typically receives a desired end-effector position and orientation in space, and it must compute the joint variables that achieve this pose. Determining these joint angles is the task of inverse kinematics.

Inverse kinematics is inherently more complex than forward kinematics, as it often involves solving nonlinear equations, handling multiple possible solutions, and ensuring that the resulting joint configurations are physically feasible. For a 6-DOF articulated robot, such as the arm considered in this research, these challenges become more pronounced due to the coupled rotational axes and the geometric constraints of the mechanism.

In this section, the inverse kinematics problem is formulated using the transformation matrix of the end-effector, derived previously. The matrix elements of the final transformation T_{ef} are now treated as known inputs, and all subscripts are redefined from f to i to emphasize their role as inverse-kinematics variables. Based on this formulation, the analytical approach for obtaining joint angles will be developed and discussed.

$$T_{ei} = \begin{bmatrix} p_{xi} \\ p_{yi} \\ p_{zi} \\ Z_{ei} \\ Y'_{ei} \\ Z''_{ei} \end{bmatrix}$$

Once the end-effector transformation matrix T_{ei} is defined, the rotational angles for each joint can be obtained by reconstructing the full transformation matrix in reverse. The translational component is directly accessible: the elements $[p_{xi}, p_{yi}, p_{zi}]$ occupy the upper portion of the final column of the matrix. To determine the rotational component, the rotation matrix must be expressed in terms of the Euler angles supplied as inputs. The rotation sequence corresponding to the $Z'Y'Z''$ Euler convention provides the necessary formulation for this process.

By substituting the inverse-kinematics Euler angles into the appropriate elemental rotation matrices, the complete rotation matrix is obtained. Throughout this derivation, the index i is appended to all variables to distinguish them as inverse-calculation parameters. These substitutions and notational conventions form the basis for constructing the rotation matrix used to compute the joint variables of the 6-DOF robot.

To systematically address the inverse kinematics problem, the robot is decomposed into two functional subsystems. This approach follows the conventional separation of an articulated manipulator into its **positioning structure** and **orientation structure**. The natural division occurs at the spherical wrist, where the final three joint axes intersect at a single point. By isolating the wrist center, the first three joints are used to determine the position of this point, while the remaining three joints are responsible for achieving the desired end-effector orientation. Based on this decomposition, the inverse kinematics solution is derived sequentially. The joint variables associated with the positioning subsystem are solved first, starting with θ_{01} , θ_{03} , and θ_{02} , as these angles directly influence the spatial placement of the wrist center. Once the wrist position is determined, the orientation subsystem is solved through θ_{05} , θ_{04} , and θ_{06} , which collectively produces the required end-effector orientation. This structured solution order ensures clarity, reduces coupling between equations, and aligns with the kinematic characteristics of the 6-DOF robot.

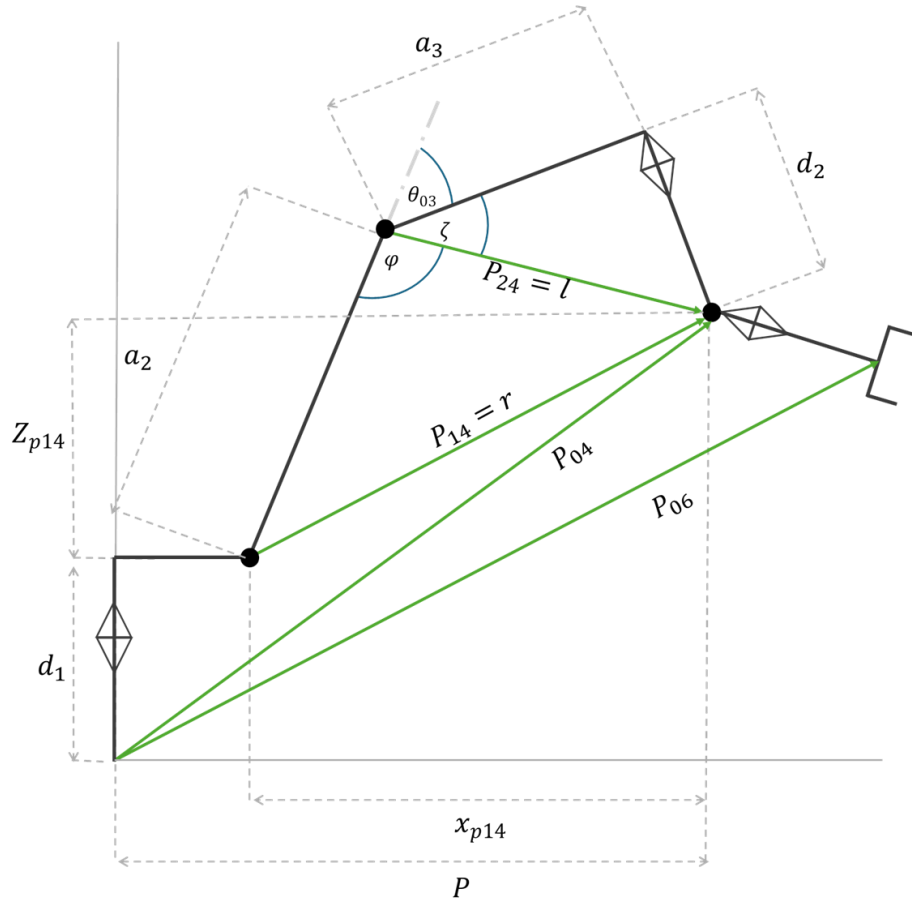


Figure 4: 2D Geometrical analysis for the robot

- For computational clarity, the following notation is adopted

$$A_{nn} = \begin{bmatrix} R_{nn,nn} & x_{nn} \\ y_{nn} \\ z_{nn} \end{bmatrix}$$

$$R_{nn,nn} = [A_{nn,rx} \ A_{nn,yx} \ A_{nn,zx}]$$

$$A_{01} = \begin{bmatrix} R_{01,11} & R_{01,12} & R_{01,13} & x_{nn} \\ R_{01,21} & R_{01,22} & R_{01,23} & y_{nn} \\ R_{01,31} & R_{01,32} & R_{01,33} & z_{nn} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{01,r} = [A_{01,rx} \ A_{01,yx} \ A_{01,zx}]$$

$$P_{nn} = \begin{bmatrix} x_{Pnn} \\ y_{Pnn} \\ z_{Pnn} \end{bmatrix}$$

1. Solving for θ_{01}

The wrist center position P_{04} is determined by deriving the input vectors from the given end-effector transformation T_{ei} , extract the translation $P_{06} = [p_{xi}, p_{yi}, p_{zi}]^T$ and the end-effector z-axis unit vector $a = [a_{xi}, a_{yi}, a_{zi}]^T$. The wrist centre P_{04} (the origin of joint 4) is obtained by subtracting the end-effector offset along a :

$$P_{04} = P_{06} - P_{46}$$

$$P_{04} = \begin{bmatrix} x_{P04} \\ y_{P04} \\ z_{P04} \end{bmatrix} = \begin{bmatrix} P_{xi} \\ P_{yi} \\ P_{zi} \end{bmatrix} - d_{6i} \begin{bmatrix} r_{13} \\ r_{23} \\ r_{33} \end{bmatrix} = \begin{bmatrix} P_{xi} - d_{6i}r_{13} \\ P_{yi} - d_{6i}r_{23} \\ P_{zi} - d_{6i}r_{33} \end{bmatrix}$$

Subsequently, the angle of P_{04} relative to the XY-plane is calculated as

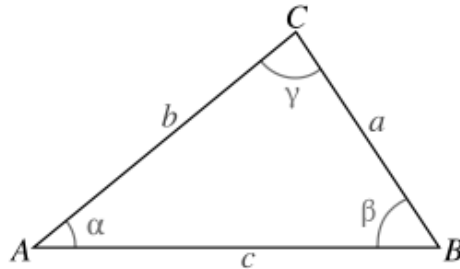
$$\theta_{01} = \theta_{01v} = \text{atan2} \left(\frac{P_{yi} - d_{6i}r_{23}}{P_{xi} - d_{6i}r_{13}} \right) = \text{atan2}(y_{P04}, x_{P04})$$

2. Solving for θ_{03}

Among all six joint variables, θ_{03} is the most challenging to determine. Once the geometric reasoning behind θ_{03} is understood, the remainder of the inverse-kinematics solution becomes significantly more straightforward. To simplify this part of the geometry, a practical constraint is introduced on the allowable range of θ_{03} , which reduces ambiguity and ensures a consistent solution for the robot configuration.

$$\theta_{03} \geq 0$$

Based on law of cosines



$$\cos(\alpha) = \frac{c^2 + b^2 - a^2}{2 \times a \times c}$$

From the geometric projection in Figure 4, the following parameters are derived

- For P_{04} projection on the XY – Plane ::

$$P = \sqrt{x_{P04}^2 + y_{P04}^2}$$

- For P_{14} projections:

$$x_{p14} = P - a_1$$

$$z_{p14} = z_{p04} - d_1$$

- Calculation of the radial distance r :

$$r = \sqrt{x_{p14}^2 + z_{p14}^2}$$

- Finding the l :

$$l = \sqrt{a_3^2 + d_2^2}$$

- Using law of cosines:

$$c = \frac{a_2^2 + l^2 + r^2}{2 \times a_2 \times l}$$

$$\varphi = \arccos(c)$$

- Calculation of the fixed shoulder offset angle (ζ):

$$\zeta = \text{atan2}(d_2, a_3) = 83.8010^\circ$$

- Constraining $\theta_3 \geq 0$ for the elbow-up configuration:

$$\theta_{03} = 180 - (\varphi + \zeta)$$

3. Solving for θ_{02}

- The angle θ_2 is decomposed into two components, β_{01} and β_{02}

- Finding beta 01

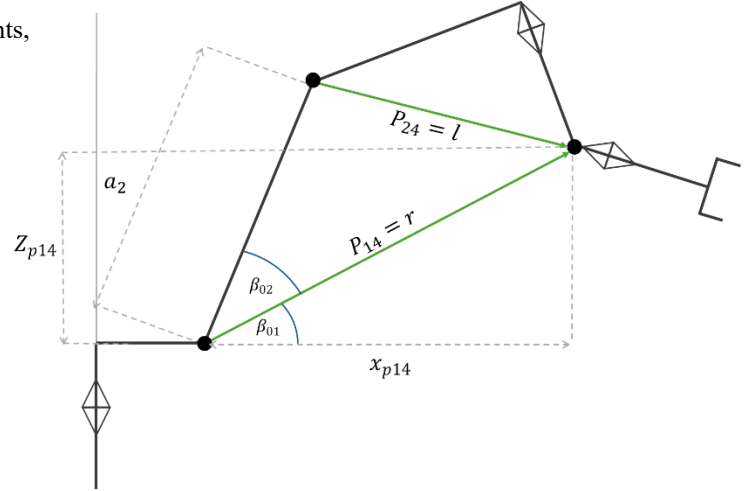
$$\beta_{01} = \text{atan2}(z_{p14}, x_{p14})$$

- Finding beta 02

$$\beta_{02} = \arccos\left(\frac{a_2^2 + r^2 + l^2}{2 \times a_2 \times r}\right)$$

- Finding theta 03

$$\theta_{02} = \beta_{01} + \beta_{02}$$



4. Solving for θ_{05}

The joint angle θ_5 is derived from the angle between the Z-axis of joint 4 and the z direction of joint 6

- Finding Z direction of 4th joint

$$A_{1i} = \begin{bmatrix} \cos \theta_{1f} & 0 & \sin \theta_{1f} & a_{1f} \cos \theta_{1f} \\ \sin \theta_{1f} & 0 & -\cos \theta_{1f} & a_{1f} \sin \theta_{1f} \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = A_{2i} = \begin{bmatrix} \cos \theta_{2f} & \sin \theta_{2f} & 0 & a_{2f} \cos \theta_{2f} \\ \sin \theta_{2f} & \cos \theta_{2f} & 0 & a_{2f} \sin \theta_{2f} \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = A_{3i} = \begin{bmatrix} \cos \theta_{3f} & 0 & -\sin \theta_{3f} & a_{3f} \cos \theta_{3f} \\ \sin \theta_{3f} & 0 & \cos \theta_{3f} & a_{3f} \sin \theta_{3f} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_{03i} = A_{1i} \times A_{2i} \times A_{3i}$$

- So $z_{03} =$

$$A_{03,rz} = \begin{bmatrix} c_1(s_2c_2 - c_2s_3) \\ s_1(s_2c_2 - c_2s_3) \\ -s_{23} \end{bmatrix}$$

- And z_{06}

$$A_{06,rz} = \begin{bmatrix} r_{13} \\ r_{23} \\ r_{33} \end{bmatrix}$$

- Finding theta 05

$$\theta_{05} = \arccos\left(\frac{A_{06,rz} \cdot A_{03,rz}}{|A_{06,rz}| |A_{03,rz}|}\right)$$

5. Solving for θ_{04}

$$A_{46i} = A_{13i}^{-1} \times A_{06i} = \begin{bmatrix} I_x & J_x & K_x & L_x \\ I_y & J_y & K_y & L_y \\ I_z & J_z & K_z & L_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- On the other hand

$$A_{46i} = A_4 \times A_5 \times A_6 = \begin{bmatrix} E_x & F_x & G_x & H_x \\ E_y & F_y & G_y & H_y \\ E_z & F_z & G_z & H_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- | | |
|-------------------------------|------------------------|
| ○ $E_x = c_4c_5c_6 - s_4s_6$ | ○ $G_x = -c_4s_5$ |
| ○ $E_y = c_4s_6 + c_5c_6c_4$ | ○ $G_y = -s_4s_5$ |
| ○ $E_z = c_6s_5$ | ○ $G_z = c_5$ |
| ○ $F_x = -c_6s_4 - c_4c_5s_6$ | ○ $H_x = -d_3c_4s_5$ |
| ○ $F_y = c_4c_6 - c_5s_4s_6$ | ○ $H_y = -d_3s_4s_5$ |
| ○ $F_z = -s_5s_6$ | ○ $H_z = d_2 + d_3c_5$ |

- For theta 04

$$\theta_{04} = \text{atan2}\left(\frac{L_y}{L_x}\right) + 180^\circ$$

6. Solving for θ_{06}

$$\theta_{06} = -\text{atan2}\left(\frac{J_z}{I_z}\right)$$

5. CONCLUSION

This research presented a comprehensive analytical derivation of forward and inverse kinematics for a 6-DOF industrial manipulator featuring joint offsets and a spherical wrist. By employing the standard Denavit-Hartenberg (DH) convention, a forward kinematic model was established to successfully map joint space to Cartesian task space.

For the inverse kinematics, a geometric decomposition approach was utilized. This method effectively decoupled the positioning problem (joints 1–3) from the orientation problem (joints 4–6) by isolating the wrist center. This geometric strategy proved advantageous over algebraic or numerical iterative methods, offering a deterministic solution that accounts for the specific mechanical offsets found in KUKA-style architecture.

The resulting mathematical model provides a computationally efficient and physically interpretable framework for robot control. The validation of these equations establishes a robust foundation for high-level tasks. Future work will extend this kinematic model to implement real-time trajectory planning, singularity analysis, and dynamic control algorithms.